# LFH AI Agent

Production Deployment Guide — v1.1.0

Updated: March 10, 2026  CDN: yannicksegaar.github.io/lfh-cdn

Branch: staging → main

● WHAT CHANGED IN V1.1.0

## Release Summary

Critical bug fixes, production naming convention, documentation, and branch safety.

| 2 | 15 | 20 | 3 |
|---|---|---|---|
| BUGS FIXED | FILES RENAMED | HEADERS ADDED | DOCS CREATED |

| CHANGE | TYPE | IMPACT |
|---|---|---|
| 4 extensions imported from non-existent modal file | ● BUG FIX | Cross-navigation between modals would crash |
| Hub tours tab imported non-existent booking form | ● BUG FIX | Hub booking button would crash |
| All 15 extensions renamed to `lfh-{feature}.js` | ● CHANGED | Clean, scannable filenames without version noise |
| EXTENSIONS.md registry created | ● NEW | Single source of truth for all extensions |
| CLAUDE.md handoff doc created | ● NEW | AI agent can understand the repo instantly |
| Branch protection on `main` | ● NEW | Prevents accidental direct pushes to production |

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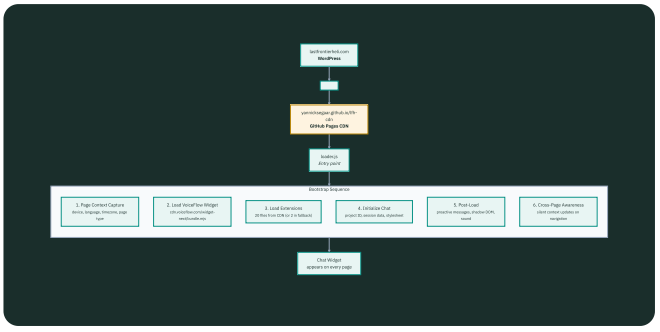
PART 1 — ARCHITECTURE OVERVIEW

## How It Works

WordPress loads one script tag. That script bootstraps the entire AI agent: context detection, VoiceFlow widget, 20 custom extensions, styling, and cross-page awareness.

**Key insight:** WordPress only needs one line of code. Everything else lives on YOUR GitHub repo. When you update something, push to GitHub — no WordPress access needed.




### The Embed Snippet

```
<!-- Last Frontier AI Chat Agent — Managed by Yannick/RomAIx -->
<script type="module"
  src="https://yannicksegaar.github.io/lfh-cdn/loader.js?v=1.1.0"
  crossorigin></script>
```

This goes in the site-wide footer, before `</body>`. Use a WordPress plugin like "WPCode" so it survives theme updates.

LOADER PIPELINE

## What loader.js Does (Step by Step)

**Context**
Device, language, timezone, page type, UTMs

**VoiceFlow**
Load widget bundle from voiceflow.com

**Extensions**
Load 20 JS files (or 2 in fallback)

**Initialize**
Start chat with session data + extensions

**Post-Load**
Proactive messages, styling, sound

**Cross-Page**
Track navigation between pages

CDN REPOSITORY CONTENTS

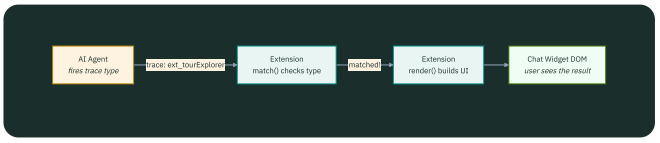
## What's Inside lfh-cdn

| FOLDER | CONTENTS | SIZE |
|---|---|---|
| `extensions/` | 20 JavaScript extension files | ~600 KB |
| `images/` | Tour photos, lodge photos, logos, backgrounds | ~130 MB |
| `fonts/` | Nexa Rust Sans Black (brand font) | ~300 KB |
| `css/` | LFH_styles.css (widget theme) | ~8 KB |
| `styles/` | Shadow DOM inject + notification sound scripts | ~16 KB |
| `audio/` | Notification beep sound | ~68 KB |
| `loader.js` | Main loader script (what WordPress loads) | ~12 KB |
| `EXTENSIONS.md` | Extension registry (source of truth) | ~4 KB |
| `CLAUDE.md` | AI agent handoff document | ~3 KB |
| `CHANGELOG.md` | Version history | ~1 KB |

PART 2 — EXTENSION REGISTRY

## All 20 Extension Files

Extensions follow the VoiceFlow pattern: `{ name, type: 'response', match, render }`. The AI agent fires trace types; extensions match and render UI.




### Exit-Triggered Extensions (fired by AI agent exit conditions)

| FILE | TRACE TYPES | OPENS MODAL | PURPOSE |
|---|---|---|---|
| `lfh-lead-form.js` | `ext_lastFrontierLeadForm_v4`, `_unified` | — | Contact / booking inquiry form |
| `lfh-tours-grid.js` | `ext_tourExplorer`, `_booking_unified` | lfh-tours-modal | Tour card grid (in-chat) |
| `lfh-lodges-widget.js` | `ext_lodgeCompare_v2`, `_unified` | lfh-lodges-modal | Lodge comparison cards (in-chat) |
| `lfh-weather-widget.js` | `ext_weather_conditions_v2`, `_unified` | lfh-weather-modal | Weather hero card (in-chat) |
| `lfh-selfservice-widget.js` | `ext_browserSelfService_v4`, `_unified` | lfh-selfservice-modal | FAQ + video browser (in-chat) |

| FILE | TRACE TYPES | OPENS MODAL | PURPOSE |
|---|---|---|---|
| `lfh-feedback.js` | `ext_feedback10` | — | Thumbs up/down after responses |

### Canvas-Triggered Extensions (fired by VoiceFlow canvas directly)

| FILE | TRACE TYPE | PURPOSE |
|---|---|---|
| `lfh-welcome.js` | `ext_welcomeGrid_v2` | Welcome screen with quick-start options |
| `lfh-snowfall.js` | `ext_snowfall1` | Decorative snowfall animation |

### Extension-Triggered (opened by other extensions)

| FILE | OPENED BY | PURPOSE |
|---|---|---|
| `lfh-hub.js` | ext_hubModal trace or other extensions | Tabbed navigation hub |
| `lfh-tours-modal.js` | tours-grid, hub, lodges, weather, selfservice | Tour detail + booking |
| `lfh-lodges-modal.js` | lodges-widget, hub, weather, selfservice | Lodge detail view |
| `lfh-weather-modal.js` | weather-widget, hub, lodges | Weather detail |
| `lfh-selfservice-modal.js` | selfservice-widget | FAQ full view |

### Data / Utility Files (imported, not registered as extensions)

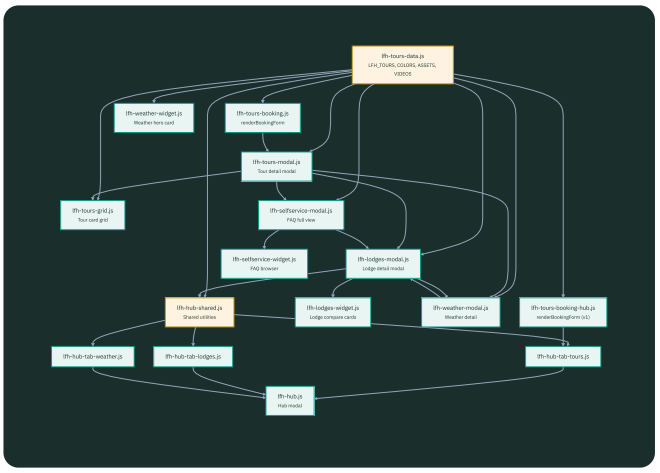
| FILE | EXPORTS | USED BY |
|---|---|---|
| `lfh-tours-data.js` | LFH_TOURS, LFH_COLORS, LFH_ASSETS, LFH_VIDEOS | Almost everything |
| `lfh-tours-booking.js` | renderBookingForm, TOUR_DATES | lfh-tours-modal |
| `lfh-tours-booking-hub.js` | renderBookingForm, TOUR_DATES | lfh-hub-tab-tours |

| FILE | EXPORTS | USED BY |
| --- | --- | --- |
| `lfh-hub-shared.js` | Colors, utilities, activity tracking | Hub tab files |
| `lfh-hub-tab-tours.js` | renderToursTab | lfh-hub |
| `lfh-hub-tab-lodges.js` | renderLodgesTab | lfh-hub |
| `lfh-hub-tab-weather.js` | renderWeatherTab | lfh-hub |

PART 3 — EXTENSION DEPENDENCY GRAPH

## How Extensions Import Each Other

Arrows mean "imports from". The core data file `lfh-tours-data.js` is the foundation — almost everything depends on it.




**Rule:** Never delete a file without checking this graph. If another file imports from it, deletion will cause a runtime crash.

PART 4 — PRE-FLIGHT VERIFICATION

## Before the Meeting with Gwyn

### 1 Verify the CDN is live

Open each URL in your browser. Each should load (not 404):

- [ ] `https://yannicksegaar.github.io/lfh-cdn/loader.js`
- [ ] `https://yannicksegaar.github.io/lfh-cdn/css/LFH_styles.css`
- [ ] `https://yannicksegaar.github.io/lfh-cdn/extensions/lfh-lead-form.js`
- [ ] `https://yannicksegaar.github.io/lfh-cdn/images/LFH_Logo_FullName_White.svg`
- [ ] `https://yannicksegaar.github.io/lfh-cdn/fonts/NexaRustSansBlack2.woff2`

### 2 Test on a local page

```
# Create a quick test page
cat > /tmp/lfh-cdn-test.html << 'EOF'
<!DOCTYPE html>
<html><head><title>LFH CDN Test</title></head>
<body>
<h1>LFH CDN Test Page</h1>
<script type="module"
  src="https://yannicksegaar.github.io/lfh-cdn/loader.js?v=1.1.0"
  crossorigin></script>
</body></html>
EOF

# Serve it (ES modules need a server, not file://)
cd /tmp && python3 -m http.server 8080

# Open: http://localhost:8080/lfh-cdn-test.html
```

### 3 Verify each extension works

- [ ] Chat widget bubble appears in bottom-right (give it 5-10 seconds)
- [ ] Proactive message: "Curious about heliskiing in Northern BC?"
- [ ] Welcome grid with quick-start options appears on open

- [ ] "Show me your tours" → tour grid appears
- [ ] "Compare the lodges" → lodge cards appear
- [ ] "What are the snow conditions?" → weather widget appears
- [ ] "I want to book a trip" → lead form eventually appears
- [ ] Thumbs up/down appears after agent responses

### 4 Check browser console

**Green Flags**

`[LFH] Extensions loaded: 9` — all extensions loaded successfully. No red errors from `lfh-cdn`.

**Red Flags**

`Failed to load module script`, `CORS error`, or `net::ERR_ABORTED 404` — a file is missing or misconfigured.

### 5 Test fallback mode

Change URL to `loader.js?v=1.1.0&fallback=true`. Only lead form + feedback should work. Tour explorer, lodges, weather, hub — these should NOT appear.

### 6 Test on mobile

- [ ] Widget bubble appears and is tappable
- [ ] Chat opens full-screen on mobile
- [ ] Extensions render correctly on small screens
- [ ] Can scroll within modals

PART 5 — GWYN MEETING SCRIPT

## Step by Step During the Meeting

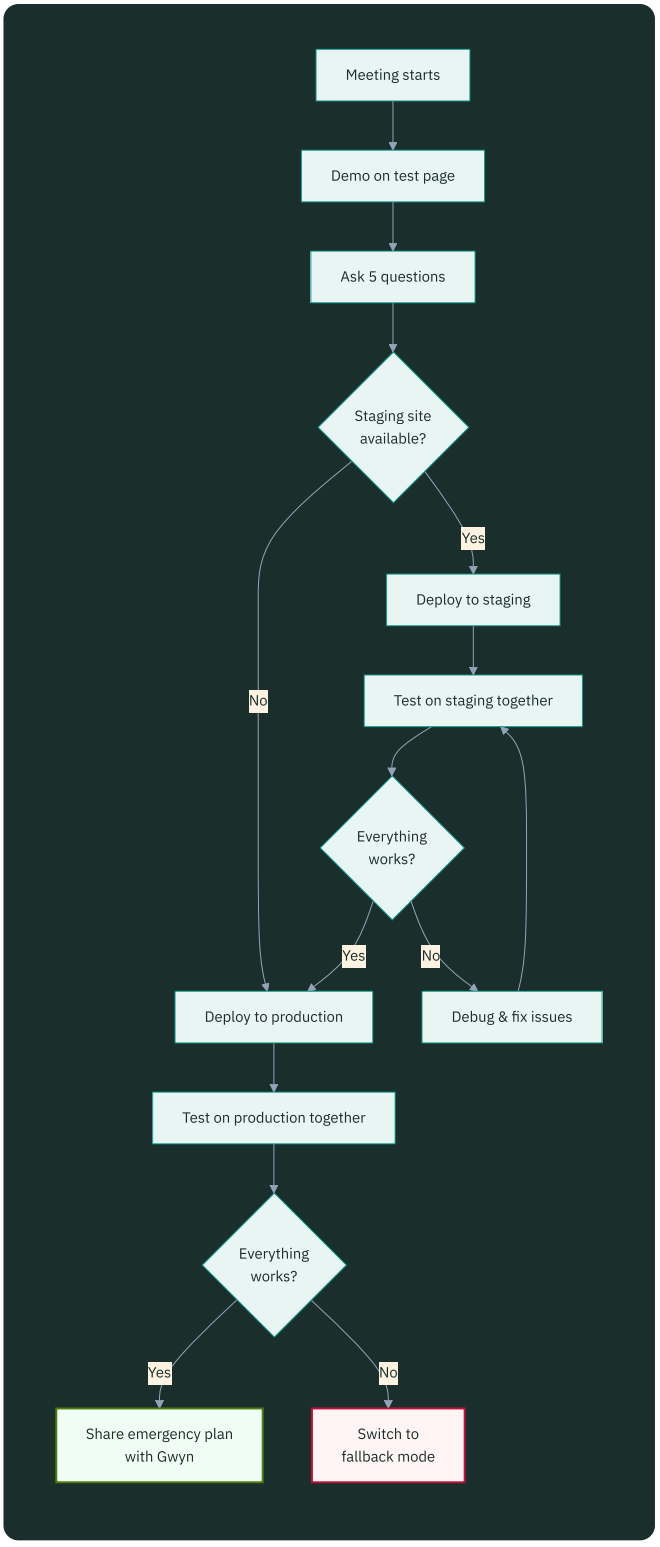
### 1 Demo first (2 minutes)

Show Gwyn the agent working on your test page. Open the chat, ask a few questions, show tour explorer, lodge comparison. Builds confidence that it works.

### 2 Ask these 5 questions (write down answers)

| QUESTION | ANSWER (FILL IN) |
|---|---|
| **URL paths:** "What are the exact URL slugs for tours, Bell 2, Ripley Creek, booking, safety, FAQ, contact, gallery, about?" | |
| **Staging site:** "Is there a staging/dev version we can test first?" | |
| **Existing chat tools:** "Are there any existing chatbots installed?" | |
| **Caching:** "Do you use WP Rocket, W3 Total Cache, or similar?" | |
| **Theme setup:** "Is this a child theme? Will the script survive updates?" | |

### 3 Deploy together




## 4 Test on production together

Visit each page and verify the widget appears:

- [ ] Home page
- [ ] Tours page

- [ ] Bell 2 Lodge page
- [ ] Ripley Creek page
- [ ] Booking / reservations page
- [ ] Contact page

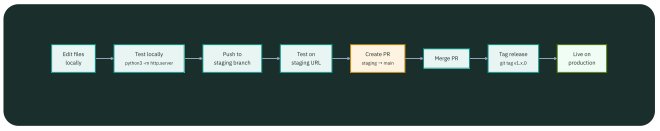
### 5 Share the emergency plan with Gwyn

**Quick fix (30 sec):** Add `&fallback=true` to the script URL → simple text-only chat.
**Remove entirely (10 sec):** Delete the script tag → chat disappears, no side effects.
**Code fixes:** I handle these by pushing to GitHub. No WordPress access needed.

PART 6 — BRANCH STRATEGY & SAFE UPDATES

## Never Push Directly to Main

The `main` branch is production — what the live website loads. It's branch-protected: direct pushes are blocked. All changes go through `staging` first.




### Step-by-Step Update Workflow

```
# 1. Switch to staging branch
cd ~/my_projects/lfh-cdn
git checkout staging
git pull origin staging

# 2. Make your changes
#    (edit extension files, loader.js, CSS, etc.)

# 3. Test locally
python3 -m http.server 8080
#    Open http://localhost:8080 and verify

# 4. Commit and push to staging
git add extensions/lfh-tours-grid.js  # (specific files, not -A)
git commit -m "Description of what changed"
git push origin staging

# 5. Test on staging URL (if configured)

# 6. Create PR to main
gh pr create --base main --head staging \
  --title "v1.x.0: Description"
```

```
# 7. Merge PR on GitHub
gh pr merge --merge

# 8. Tag the release
git checkout main && git pull
git tag v1.x.0 && git push --tags

# 9. Update CHANGELOG.md
```

**Why this matters:** Before v1.1.0, pushing to `main` went live instantly. One typo = broken widget for all visitors. The staging branch gives you a safety net to test before going live.

PART 7 — HOW TO MAKE UPDATES AFTER LAUNCH

## Common Changes & Where to Edit

| WHAT TO CHANGE | WHERE TO EDIT | NOTES |
|---|---|---|
| Proactive greeting messages | `loader.js` lines 202-213 | Change the text strings |
| Chat widget title | `loader.js` line 189 | Currently "Last Frontier" |
| Page type detection | `loader.js` lines 63-75 | Update if WordPress URLs change |
| Widget styling | `css/LFH_styles.css` | Colors, fonts, layout |
| Extension behavior | `extensions/*.js` | The interactive components |
| Background image | `images/` | Replace file, keep the name |
| Shadow DOM styling | `styles/lastfrontier-shadow-inject.js` | Chat BG, header, font |

### After the Meeting: Update Page Type Detection

Using the URL paths Gwyn gave you, update `detectPageType()` in `loader.js` (around line 63). Example:

```
// If Gwyn says tours is at /heliskiing-tours/
if (p.includes('/tour') || p.includes('/heli-ski') || p.includes('/heliskiing'))
  return 'tours';
```

### VoiceFlow Cross-Page Awareness (optional, can do later)

Already built into loader.js. To enable in VoiceFlow:

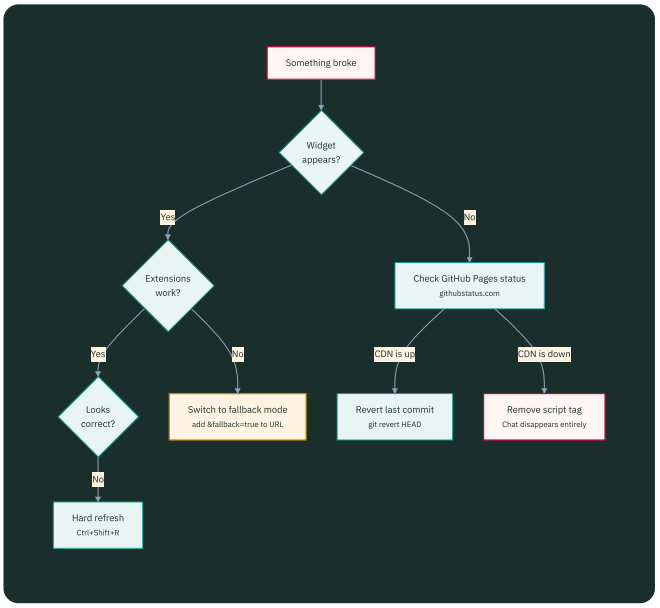
1. Go to the main Listen/Conversation node in VoiceFlow
2. Add an Event Trigger: `page_context_update`
3. Connect to a Code Step with this code:

```
var data = (last_event && last_event.payload)
  ? last_event.payload : last_event;
if (data) {
  if (data.page_type) page_type = data.page_type;
```

```
    if (data.page_path) page_path = data.page_path;
  }
```

PART 8 — EMERGENCY PROCEDURES

# When Something Goes Wrong




## Response Options (fastest first)

### Fallback Mode (30 sec)

Tell Gwyn: "Add `&fallback=true` to the script URL." Result: simple text chat + lead form only.

### Roll Back Code (2 min)

`git revert HEAD && git push origin main` — undoes the last commit. Wait 2-3 min for GitHub Pages.

### Remove Entirely (10 sec)

Tell Gwyn: "Delete the script tag from the footer." Chat disappears. No side effects on the website.

## Key URLs

| WHAT | URL |
|---|---|
| CDN root | `https://yannicksegaar.github.io/lfh-cdn/` |
| Loader script | `https://yannicksegaar.github.io/lfh-cdn/loader.js` |
| GitHub repo | `https://github.com/YannickSegaar/lfh-cdn` |
| VoiceFlow project | Dashboard → Project ID: `699ffcecd1c4a9bd2986b463` |
| GitHub Pages settings | `github.com/YannickSegaar/lfh-cdn/settings/pages` |

## Naming Convention

**Format:** `lfh-{feature}.js` — no version numbers, no "unified" suffixes.

Version tracked in git tags ( `v1.1.0` ) and file header comments. Each file has a header documenting its origin, trace types, and dependencies.

## Origin Map (dev → prod)

| PRODUCTION NAME | DEV ORIGIN |
|---|---|
| `lfh-lead-form.js` | `lead-capture-form-v5-unified.js` |
| `lfh-feedback.js` | `lfh-feedback-v10.js` |
| `lfh-welcome.js` | `welcome-grid-v2-unified.js` |
| `lfh-snowfall.js` | `snowfall.js` |
| `lfh-tours-grid.js` | `lfh-tour-explorer-grid-booking-unified.js` |
| `lfh-tours-modal.js` | `lfh-tour-explorer-modal-booking-unified-v2.js` |
| `lfh-tours-data.js` | `lfh-tour-explorer-modal.js` |
| `lfh-tours-booking.js` | `lfh-tour-booking-form-v2.js` |
| `lfh-tours-booking-hub.js` | `lfh-tour-booking-form.js` (v1) |

| PRODUCTION NAME | DEV ORIGIN |
| --- | --- |
| `lfh-lodges-widget.js` | `lfh-lodge-compare-widget-v2-unified.js` |
| `lfh-lodges-modal.js` | `lfh-lodge-compare-modal-v2-unified.js` |
| `lfh-weather-widget.js` | `lfh-weather-conditions-widget-v2-unified.js` |
| `lfh-weather-modal.js` | `lfh-weather-conditions-modal-unified.js` |
| `lfh-selfservice-widget.js` | `browser-self-service-v4-widget-unified.js` |
| `lfh-selfservice-modal.js` | `browser-self-service-v4-modal-unified.js` |
| `lfh-hub.js` | `lfh-hub-modal.js` |

## Fallback Mode Comparison

| FEATURE | FULL MODE | FALLBACK MODE |
| --- | --- | --- |
| Text chat + KB answers | Yes | Yes |
| Lead capture form | Yes | Yes |
| Feedback (thumbs) | Yes | Yes |
| Tour explorer | Yes | No |
| Lodge comparison | Yes | No |
| Weather conditions | Yes | No |
| Self-service browser | Yes | No |
| Hub modal | Yes | No |
| Welcome grid | Yes | No |
| Snowfall | Yes | No |
| Cross-page awareness | Yes | Yes |

QUICK REFERENCE CARD

## Copy-Paste Commands

### The WordPress Embed Line

```
<script type="module"
  src="https://yannicksegaar.github.io/lfh-cdn/loader.js?v=1.1.0"
  crossorigin></script>
```

### Switch to Fallback

```
<script type="module"
  src="https://yannicksegaar.github.io/lfh-cdn/loader.js?v=1.1.0&fallback=true"
  crossorigin></script>
```

### Push an Update (safe workflow)

```
cd ~/my_projects/lfh-cdn
git checkout staging
# make changes, test locally
git add <files> && git commit -m "what changed"
git push origin staging
gh pr create --base main --head staging --title "v1.x.0: what changed"
# merge PR on GitHub, then:
git checkout main && git pull
git tag v1.x.0 && git push --tags
```

### Roll Back

```
cd ~/my_projects/lfh-cdn
git checkout main
git revert HEAD
git push origin main  # wait 2-3 min for GitHub Pages
```